

Reflection Essay Draft

The 2026 winter and spring capstone senior project has provided opportunities to develop skills relating to an engineering career. Working on the Sumobot project, skills have been developed which closely align with the National Association of Colleges and Employers competencies for a career-ready workforce. My Sumobot project has allowed me to exercise teamwork, critical thinking, technology, communication, and self-development.

Teamwork has been a foundational aspect of the Sumobot project. Being able to work effectively within the team by building comradery and trust has contributed heavily to the success of the project. Specifically, I have built trust by entrusting others to develop important pieces of the Sumobot such as offensive mechanisms which are important to the functionality, and competitive aspect of the robot. Building comradery within the team has also allowed me to meet up outside of class time, and at group members houses to work on the project. This has allowed me and the team to work efficiently and get tasks done faster. Being able to build comradery and trust with peers in a team will be an extremely valuable skill when it comes to interview success, longstanding work relationships, and job security

Because the Sumobot project involves a final competition, we have had to use our critical thinking skills and design around other robots for competitive advantages. Critical thinking has been applied with our project through design iterations. We have had to make improvements to prototype versions, constantly creating new designs and problem-solving issues. I applied critical thinking skills when creating an improved servo deployment arm which better fits the needs of our robot. Shape, material usage, material weight, and maximizing space efficiency were all used in the critical thinking process. I will take this skill of critical thinking to the engineering industry, utilizing it to add valuable design choices to products, fix existing design problems, and be a valuable overall problem-solver.

Technology is an extremely important skill in an engineering career. Technology will be used in almost every aspect of the engineering process including the creation, documentation, and organization of team efforts. I used technology in this product for the creation of the robot frame using 3D CAD modeling. I used OnShape to model and dimension the first frame iteration, various ramp mechanisms, and the servo deployment arm. I also used technology such as a 3D printer to quickly manufacture these parts for rapid prototyping. Using technology to rapidly produce prototypes and to better convey design ideas and strategies is a mandatory competency in the engineering workforce.

Over the course of the Sumobot project, communication skills have also been heavily developed. Communication between team members has been essential to maintaining project organization and ensuring that all members maintained a predictable workflow. Communication was used during meetings, brainstorming sessions, design discussions, troubleshooting, and planning meetup times. I contributed to communication efforts by discussing design improvements, coordinating manufacturing tasks, and explaining design concepts to teammates. Communication has also been important through project documentation and presentations. Being able to effectively communicate technical information is an important engineering skill because engineers must regularly explain ideas, designs, and problems to coworkers, managers, and clients in a professional environment.

The Sumobot project has also contributed to self-development throughout the capstone process. During the project, I have learned how to better manage time, adapt to unexpected problems, and improve technical skills independently. There were many situations where design issues or manufacturing failures required additional research and learning outside of normal class instruction. For example, troubleshooting 3D printing issues, redesigning components to better fit within size constraints, and improving the servo arm design all required self-motivation and the willingness to continue improving previous work. Working on a long-term engineering project has helped me become more confident in my technical abilities and problem-solving skills. Self-development is important in engineering because technology and design methods are constantly changing, requiring engineers to continuously learn and adapt throughout their careers.

Overall, the Sumobot capstone project has provided valuable experience that directly relates to an engineering career and the National Association of Colleges and Employers career readiness competencies. The project has strengthened my abilities in teamwork, critical thinking, technology, communication, and self-development through hands-on engineering experience. These skills will help prepare me for future opportunities in the industry by allowing me to contribute effectively in technical teams, solve engineering problems, communicate professionally, and continue improving as an engineer.